A Pandrosion Operator for Derivative-Free Polynomial Root-Finding

Universal polynomial evaluation, non-local geometry, and an algorithmic companion to Smale's 17th problem

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Abstract

We extend the generalized Pandrosion iteration — a geometric fixed-point method originally designed for computing p-th roots via the geometric progression polynomial $S_p(z)$ — from the real line to the complex plane and, crucially, to *polynomial systems* $F : \mathbb{C}^n \to \mathbb{C}^n$, the proper setting of Smale's 17th problem (1998).

The univariate Pandrosion sequence is the special case $n = 1$ of a *universal evaluation generator*: by anchoring the matrix-valued difference quotient $\mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})$ defined by the telescoping identity $F(\mathbf{z}) - F(\mathbf{z}_0) = \mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})\,(\mathbf{z} - \mathbf{z}_0)$, we define a non-local rational iteration $\mathbf{z}_{n+1} = \mathbf{z}_0 - \mathbf{Q}_F(\mathbf{z}_0, \mathbf{z}_n)^{-1} F(\mathbf{z}_0)$ that recovers the exact p-th root formula in $n = 1$, the multivariate Newton iteration as the limiting case where the anchor tracks the iterate, and applies uniformly to any polynomial map F.

For univariate $P(z) = xz^p - 1$ we characterize the complex contraction ratio.

Our main rigorous contributions, in both univariate ($n = 1$) and multivariate ($n \geq 2$) settings, are threefold: (i) a universal local convergence theorem establishing an explicit contraction basin of radius $\eta_F(\boldsymbol{\zeta})$ around every regular zero $\boldsymbol{\zeta}$ of F; (ii) exact polynomial factorisations avoiding the singular Σ barriers of Newton's method completely; (iii) a macroscopic kinematic identity yielding a bounded telescopic product.

We identify a *far-anchor obstruction* for fixed-anchor Cauchy-circle starts and resolve it via the **scaling principle**: adaptive anchor reanchoring every $K = O(1)$ steps keeps the effective argument $\|\mathbf{Q}_F^{-1} F\| \approx 1$.

The resulting adaptive Pandrosion-T_3 scheme is verified numerically on four adversarial univariate families up to degree $d = 500$, and on multivariate Smale-17-style polynomial systems with Bézout degree d^n up to $(n, d) \in \{(2, 3), (3, 3), (4, 2)\}$ (cf. `latex/scripts/benchmark_smale17_v4.py`).

Conditionally on Conjecture 5.19, this yields a deterministic, derivative-free $O(d^2 \log d)$-operation BSS algorithm for univariate root-finding; the multivariate Smale-17 average-case $O(\mathrm{poly}(d, n))$ bound is established unconditionally only by Beltrán–Pardo [4] and Lairez [5], and our scheme provides a derivative-free *algorithmic alternative* with comparable empirical complexity, not a competing complexity-theoretic resolution. We close with a formal Pandrosion-language rewriting of the Euler product for $\zeta(s)$.

Reader's guide. This is the first companion paper after the verified monomial paper. It starts the *algorithmic* branch of the corpus: the difference-quotient operator, adaptive anchors, and multivariate lift introduced here are the objects that Parts VII–VIII later modify with fallbacks and start strategies. Parts II–VI are a separate MVC branch; they use the same quotient notation but address a different Smale problem and should not be read as prerequisites for the algorithmic complexity statements below.

Distinction from Smale's Mean Value Conjecture. The present work and the companion Parts II–VI [16] address *two distinct conjectures* of Smale, often confused in the literature:

- **Smale's Mean Value Conjecture (MVC, 1981, [17]):** a univariate inequality $\min_{P'(\zeta)=0} |P(\zeta)/(\zeta P'($ $(d-1)/d$ for monic polynomials P of degree d with $P(0) = 0$. Treated in Parts II–VI.
- **Smale's 17th Problem (1998, [18]):** a complexity-theoretic problem about the average-case cost of finding an approximate zero of a *system* $F : \mathbb{C}^n \to \mathbb{C}^n$ of n polynomials in n unknowns, with Bézout degree d^n. Resolved positively by Beltrán–Pardo (2009, [4]) probabilistically and Lairez (2017, [5]) deterministically. Treated in this Part I and in Parts VII–VIII.

The present paper develops the algorithmic framework for the latter; we do *not* claim to improve upon the worst-case complexity bound of [5], but provide a derivative-free alternative (cost only one polynomial-evaluation oracle per step, not Jacobian inverse) with explicit per-orbit constants and empirical evidence on multivariate systems matching the multivariate Pandrosion operator $\mathbf{Q}_F$ defined in §3.5.

Contents

1 Introduction

In Paper 0, we studied the generalized Pandrosion iteration

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k, \qquad v_n = x \cdot s_n^{p-1}, \tag{1}$$

for real $x > 0$ and integer $p \geq 2$. That construction, rooted in Pandrosion of Alexandria's geometric method for duplicating the cube [6], converges linearly to $s^* = x^{-1/p}$ with an explicit contraction ratio $\lambda_{p,x}$ depending only on $\alpha = x^{1/p}$. Steffensen's acceleration transforms the linear method into a quadratically convergent, derivative-free algorithm.

A natural question arises: *does Pandrosion's iteration extend to* $\mathbb{C}$*?* Nothing in the formula requires x or s to be real — the geometric sum S_p and the rational map $h(s) = 1 - (x-1)/(x \cdot S_p(s))$ are well-defined for complex arguments.

This extension is not merely formal. By moving to $\mathbb{C}$, we expose three structural features:

- **Uniform anchor-based evaluation**: Pandrosion's iteration is a special case of an anchor-based difference-quotient scheme applicable to any polynomial in $\mathbb{C}[z]$, and lifts formally to polynomial systems via a matrix-valued quotient $\mathbf{Q}_F$.

  This provides a derivative-free algorithmic framework of independent interest, and an algorithmic companion to — *not* a competing complexity-theoretic resolution of — Smale's 17th problem (already resolved by Beltrán–Pardo [4] and Lairez [5]).

- **Fractal boundary structure**: the induced complex dynamics exhibit self-similar basin boundaries distinct from those of Newton's method. We describe this structure qualitatively and do not claim new results on complex dynamics itself.

- **A structural rewriting of the Euler product**: the infinite geometric sum $S_\infty(s) = (1-s)^{-1}$ recasts the Euler factors of $\zeta(s)$ as "infinite Pandrosion sums". This is a notational observation and implies *no* consequence for the Riemann Hypothesis.

Outline. Section 2 recalls the complex Pandrosion iteration and its contraction ratio (established in [3]). Section 3 presents the universal polynomial generalization. Section 4 introduces the adaptive-anchor algorithm. Section 5 develops the rigorous convergence theory. Section 6 collects supplementary results on basins, Steffensen acceleration, the Euler connection, and the divergence boundary. Section 7 concludes.

2 Complex Pandrosion iteration: recap

Definition 2.1 (Complex Pandrosion iteration)**.** *Let $x \in \mathbb{C} \setminus \{0,1\}$ and $p \geq 2$. The complex Pandrosion iteration is*

$$h(s) = 1 - \frac{x-1}{x \cdot S_p(s)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k. \tag{2}$$

Its p fixed points are $s_k^* = \alpha_k^{-1}$, where $\alpha_k = x^{1/p} e^{2\pi i k/p}$, $k = 0, \ldots, p-1$.

Theorem 2.2 (Complex contraction ratio)**.** *At the principal fixed point $s_0^* = \alpha^{-1}$ ($\alpha = x^{1/p}$), the contraction ratio is*

$$\lambda_{p,x} = h'(s_0^*) = \frac{(\alpha - 1)\sum_{k=1}^{p-1} k\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}, \qquad \alpha = x^{1/p} \in \mathbb{C}. \tag{3}$$

The convergence region is $\mathcal{C}_p = \{x \in \mathbb{C} : |\lambda_{p,x}| < 1\}$; the divergence set $\mathcal{D}_p = \mathbb{C} \setminus \mathcal{C}_p$ is confined to a neighborhood of the negative real axis.

3 Universal polynomial generalization

The classical Pandrosion map was derived for p-th roots ($s^* = x^{-1/p}$). We now show that it is a special case of a general root-finding scheme applicable to *any* polynomial $P(z) \in \mathbb{C}[z]$.

3.1 The difference quotient operator

Let $P(z) = 0$ be a polynomial equation with unknown root z^*. Fix an anchor point $z_0 \in \mathbb{C}$ and write the polynomial remainder factorisation:

$$P(z) - P(z_0) = (z - z_0) \cdot Q(z_0, z), \tag{4}$$

where $Q(z_0, z) = \sum_{k=0}^{d-1} c_k(z_0) z^k$ is the *difference quotient polynomial* of degree $d - 1$. Since $P(z^*) = 0$, we obtain $z^* = z_0 - P(z_0)/Q(z_0, z^*)$, yielding the fixed-point iteration:

$$\boxed{z_{n+1} = \mathcal{P}_{P,z_0}(z_n) = z_0 - \frac{P(z_0)}{Q(z_0, z_n)}.} \tag{5}$$

We call $\mathcal{P}_{P,z_0}$ the *Generalized Pandrosion Operator.*

3.2 Recovering the classical iteration

For $P(s) = xs^p - 1$ with anchor $z_0 = 1$, we have $P(1) = x - 1$ and $P(s) - P(1) = x(s^p - 1) = x(s - 1)S_p(s)$, so $Q(1, s) = x \cdot S_p(s)$. The operator gives exactly

$$s_{n+1} = 1 - \frac{x - 1}{x \cdot S_p(s_n)},$$

the classical Pandrosion iteration.

3.3 Newton as a limiting case

When the anchor coincides with the current iterate ($z_0 = z_n$), the difference quotient degenerates to the derivative:

$$Q(z_n, z_n) = \lim_{z \to z_n} \frac{P(z) - P(z_n)}{z - z_n} = P'(z_n).$$

The operator reduces to $z_{n+1} = z_n - P(z_n)/P'(z_n)$, which is exactly Newton's method. Thus Newton is the Pandrosion operator with a *dynamic anchor* $z_0 = z_n$, updated at every step.

3.4 The fixed-anchor obstruction

With a fixed anchor z_0, the operator has poles wherever $Q(z_0, z_n) = 0$. Numerical measurement on a 1000×1000 grid over $[-2, 2]^2$ confirms: the fixed-anchor Steffensen-Pandrosion converges for only 38.6% of starting points on $z^3 - 1$, compared to 100% for Newton.

3.5 Multivariate extension to systems $F : \mathbb{C}^n \to \mathbb{C}^n$

The framework of this paper is *not restricted to univariate polynomials.* The construction extends verbatim to systems $F = (F_1, \ldots, F_n) : \mathbb{C}^n \to \mathbb{C}^n$ of n polynomials in n unknowns — the proper setting of Smale's 17th problem (1998, [18, 4, 5]).

Acknowledgment of classical prior art. The matrix-valued divided difference $\mathbf{Q}_F$ that we use is *not new*. It is the *Schmidt slope matrix* (also called the *Steffensen divided difference operator for systems*), introduced in [21] and developed systematically in [22, Chapter 11]. The corresponding fixed-point iteration $\mathbf{z}_{n+1} = \mathbf{z}_0 - \mathbf{Q}_F^{-1}F(\mathbf{z}_0)$ is the classical *multivariate secant method* when the anchor is fixed, and reduces to multivariate Newton when the anchor tracks the iterate. Recent derivative-free methods exploiting variants of this slope matrix include Grau-Sánchez–Noguera–Amat [24] (Ostrowski-type), Amat–Busquier [25] (Steffensen-Newton hybrids), and the textbook treatment of Potra–Pták [23]. Our *specific* contribution here is not the slope matrix itself, but its combination with: (i) the *periodic adaptive reanchoring* of Definition 3.3 (the scaling principle borrowed from [1]), (ii) the higher-order Aitken-Steffensen acceleration T_3, and (iii) the non-holomorphic fallback of [14]. We restate Schmidt's slope matrix below for self-containment.

Definition 3.1 (Schmidt slope matrix; multivariate Pandrosion difference quotient)**.** *Let $F : \mathbb{C}^n \to \mathbb{C}^n$ be a polynomial map and $\mathbf{z}_0, \mathbf{z} \in \mathbb{C}^n$ with $\mathbf{z} \neq \mathbf{z}_0$. The matrix-valued Pandrosion difference quotient is the $n \times n$ complex matrix $\mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})$ defined by the telescoping identity*

$$F(\mathbf{z}) - F(\mathbf{z}_0) \;=\; \mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})\,(\mathbf{z} - \mathbf{z}_0), \tag{6}$$

constructed entry-by-entry by interpolating between $\mathbf{z}_0$ and $\mathbf{z}$ one coordinate at a time:

$$\big[\mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})\big]_{i,j} \;=\; \frac{F_i\big(\mathbf{z}_0[0..j-1]\,|\,\mathbf{z}[j]\,|\,\mathbf{z}[j+1..]\big) - F_i\big(\mathbf{z}_0[0..j-1]\,|\,\mathbf{z}_0[j]\,|\,\mathbf{z}[j+1..]\big)}{\mathbf{z}[j] - \mathbf{z}_0[j]}, \tag{7}$$

with the convention $[\mathbf{Q}_F]_{i,j} = (\partial F_i/\partial z_j)(\mathbf{z}^{(j)})$ when the denominator vanishes (smooth limit).

Remark 3.2 (Structural meaning of $\mathbf{Q}_F$ and its history)**.** *The matrix $\mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})$ is the multivariate generalisation of the univariate divided difference $Q(z_0, z) = (P(z) - P(z_0))/(z - z_0)$, originally constructed by Schmidt [21] for the analysis of the multivariate secant method. It satisfies $\mathbf{Q}_F(\mathbf{z}, \mathbf{z}) = DF(\mathbf{z})$ (the Jacobian of F at $\mathbf{z}$) in the diagonal limit, recovering Newton's method as the dynamic-anchor case (cf. [22, Theorem 11.2.5]). Each entry of $\mathbf{Q}_F$ requires two evaluations of one component of F, hence the total cost of computing $\mathbf{Q}_F$ is $O(n^2)$ scalar polynomial evaluations, identical to the cost of evaluating the Jacobian by finite differences but without requiring the analytic Jacobian itself — the scheme is fully derivative-free.*

Definition 3.3 (Adaptive reanchoring; restated for the multivariate case)**.** *The multivariate adaptive Pandrosion-T_3 scheme alternates: (a) $K = 3$ Steffensen-accelerated steps of $\mathcal{P}_{F,\mathbf{z}_0}$ from a fixed anchor $\mathbf{z}_0$, then (b) reanchor $\mathbf{z}_0 \leftarrow \mathbf{z}_K$. The novelty is the period-K reanchoring, not the operator $\mathcal{P}_{F,\mathbf{z}_0}$ itself, which is classical (cf. above).*

Definition 3.4 (Multivariate Pandrosion operator)**.** *For a polynomial map $F : \mathbb{C}^n \to \mathbb{C}^n$ and an anchor $\mathbf{z}_0 \in \mathbb{C}^n$, the Multivariate Generalized Pandrosion Operator is*

$$\boxed{\mathcal{P}_{F,\mathbf{z}_0}(\mathbf{z}) \;=\; \mathbf{z}_0 - \mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})^{-1}\,F(\mathbf{z}_0),} \tag{8}$$

defined whenever $\mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})$ is invertible.

Theorem 3.5 (Multivariate Pandrosion fixes regular zeros)**.** *Let $\boldsymbol{\zeta} \in \mathbb{C}^n$ be a regular zero of F (i.e., $F(\boldsymbol{\zeta}) = 0$ and $\det DF(\boldsymbol{\zeta}) \neq 0$). For every anchor $\mathbf{z}_0$ in a sufficiently small neighbourhood $\mathcal{U}$ of $\boldsymbol{\zeta}$ and every $\mathbf{z} \in \mathcal{U}$, the operator $\mathcal{P}_{F,\mathbf{z}_0}(\mathbf{z})$ is well-defined and $\mathcal{P}_{F,\mathbf{z}_0}(\boldsymbol{\zeta}) = \boldsymbol{\zeta}$.*

Proof. By continuity, $\mathbf{Q}_F$ is invertible in a neighbourhood of $\boldsymbol{\zeta}$ since $\mathbf{Q}_F(\boldsymbol{\zeta}, \boldsymbol{\zeta}) = DF(\boldsymbol{\zeta})$ is invertible by hypothesis. The fixed-point property follows from $F(\boldsymbol{\zeta}) = 0$: substituting $\mathbf{z} = \boldsymbol{\zeta}$ into (6) gives $-F(\mathbf{z}_0) = \mathbf{Q}_F(\mathbf{z}_0, \boldsymbol{\zeta})\,(\boldsymbol{\zeta} - \mathbf{z}_0)$, hence $\mathcal{P}_{F,\mathbf{z}_0}(\boldsymbol{\zeta}) = \mathbf{z}_0 + (\boldsymbol{\zeta} - \mathbf{z}_0) = \boldsymbol{\zeta}$. □

Proposition 3.6 (Empirical observation, multivariate adaptive Pandrosion-T_3)**.** *The multivariate adaptive Pandrosion-T_3 scheme of Definition 3.4, with anchor period $K = 3$ and Steffensen acceleration, was tested on random polynomial systems $F : \mathbb{C}^n \to \mathbb{C}^n$ from the unitarily-invariant Kostlan–Smale ensemble (cf. Beltrán–Pardo [4]) for the configurations*

$$(n, d) \in \{(2, 3), (3, 3), (4, 2), (4, 3), (5, 2), (5, 3), (5, 4), (6, 2), (6, 3)\},$$

covering Bézout degrees $D = d^n$ from 9 up to 1024. The scheme achieves 5/5 convergence at every tested configuration (45/45 = 100% overall) with mean iteration counts:

(n, d)	D	*success*	*avg. iters*	(n, d)	D	*success*	*avg. iters*
$(2, 3)$	9	5/5	9.8	$(5, 3)$	243	5/5	24.2
$(3, 3)$	27	5/5	12.0	$(5, 4)$	1024	5/5	18.8
$(4, 2)$	16	5/5	12.0	$(6, 2)$	64	5/5	16.0
$(4, 3)$	81	5/5	19.0	$(6, 3)$	729	5/5	20.2
$(5, 2)$	32	5/5	13.4	*Mean across all configs:* 16.2			

The largest tested configuration is $(n, d) = (5, 4)$ with Bézout degree $D = 1024$ (1024 projective roots, 126 monomials per equation). This is an empirical observation, not a theorem; the convergence guarantee at the scale relevant to Smale 17 ($D \to \infty$) is open. Detailed benchmarks are in `latex/scripts/benchmark_multivariate_extended.py` *and* `latex/scripts/benchmark_smale17_v4.py`*.*

Remark 3.7 (Honest accounting of the multivariate per-orbit cost)**.** *For a polynomial system of Bézout degree $D = d^n$, evaluating F at one point costs $O(D \cdot n)$ BSS operations. The multivariate Pandrosion-T_3 uses $O(n)$ such evaluations per step (for the matrix $\mathbf{Q}_F$) and $O(n^3)$ for the linear solve $\mathbf{Q}_F^{-1} F(\mathbf{z}_0)$, giving a per-step cost of $O(nD + n^3)$. If a multivariate analogue of Theorem 5.8 could be established giving $O(\log D)$ epochs to basin entry, the per-orbit cost would be $O((nD + n^3) \log D)$. We emphasise:*

- *This $O(\log D)$ epoch bound is* ***essentially the difficulty of Smale 17 itself*** *and is not established by any of our arguments. Beltrán–Pardo [4] and Lairez [5] achieve the analogous bound via deep μ-theory and homotopy continuation respectively, methods orthogonal to ours.*

- *The per-orbit cost above must be multiplied by the number of starts (heuristically D, by analogy with the univariate case) for a system-level total.*

- *Our empirical validation extends to $(n, d) = (5, 4)$ with Bézout degree $D = 1024$, the largest setting in which the algorithm achieves 100% convergence rate; this is still well below the asymptotic regime relevant to Smale 17 ($D \to \infty$).*

Honest summary. *The multivariate Pandrosion-T_3 is a derivative-free algorithmic alternative whose practical performance matches Newton with analytic Jacobian on small-scale Smale-17-style benchmarks. Whether it admits an unconditional polynomial average-case complexity bound of the same order as Lairez [5] is an* ***open question that we do not address****.*

4 Adaptive anchor: the scaling principle in $\mathbb{C}$

The scaling optimisation of Paper 0 (Section 12) overcomes the large-x slowdown of Pandrosion's real iteration by reducing x to $x' \approx 1$. We extend this principle to the complex plane.

4.1 The adaptive Pandrosion–Steffensen algorithm

Definition 4.1 (Adaptive Pandrosion–Steffensen)**.** *Let $P \in \mathbb{C}[z]$ of degree d and $K \geq 1$ an integer. The adaptive iteration alternates:*

1. ***Pandrosion-Steffensen step**: compute $h = z_0^{(m)} - P(z_0^{(m)})/Q(z_0^{(m)}, z_n)$, then $T(z_n) = z_n - (h - z_n)^2/(h' - 2h + z_n)$ where $h' = z_0^{(m)} - P(z_0^{(m)})/Q(z_0^{(m)}, h)$, and set $z_{n+1} = T(z_n)$.*
2. ***Anchor update**: every K steps, set $z_0^{(m+1)} = z_n$.*

Remark 4.2 (Interpolation between Pandrosion and Newton)**.** *$K = \infty$: pure fixed-anchor Pandrosion. $K \to 0$: anchor tracks the iterate (Newton). $K = 3$: anchor updated every 3 steps; the denominator uses a slightly stale anchor providing non-local correction that avoids the poles of $P'(z_n)$ while staying close enough to prevent escape.*

4.2 Numerical evidence: adaptive Pandrosion surpasses Newton

Method	$z^3 - 1$	$z^5 - z - 1$
Newton	100.00%	95.51%
Fixed-anchor Pandrosion ($z_0 = 0$)	38.64%	92.16%
Adaptive Pandrosion ($K = 3$)	**100.00%**	**100.00%**

4.3 Higher-order methods

Definition 4.3 (Higher-order Pandrosion-T_n)**.** *Given $P \in \mathbb{C}[z]$, anchor z_0, and base map $h(w) = z_0 - P(z_0)/Q(z_0, w)$:*

$$T_2(z) = z - \frac{(s_1 - z)^2}{s_2 - 2s_1 + z}, \qquad \hat{\lambda} = \frac{s_2 - s_1}{s_1 - z}, \tag{9}$$

$$T_3(z) = T_2(z) - \frac{s_3 - T_2(z)}{\hat{\lambda} - 1}, \qquad s_3 = h(T_2(z)), \tag{10}$$

$$T_4(z) = T_3(z) - \frac{s_4 - T_3(z)}{\hat{\lambda} - 1}, \qquad s_4 = h(T_3(z)). \tag{11}$$

Each T_n uses exactly n evaluations of h and reaches convergence order n via Richardson extrapolation.

Proposition 4.4 (Empirical scaling law)**.** *Regression on adversarial polynomial families gives the power-law fits:*

$$n_\star^{T_2} \approx 1.68\, d^{0.70}, \tag{12}$$

$$n_\star^{T_3} \approx 1.18\, d^{0.77}, \tag{13}$$

$$n_\star^{T_4} \approx 1.27\, d^{0.70}, \tag{14}$$

$$n_\star^{\mathrm{Newton}} \approx 3.05\, d^{0.80}. \tag{15}$$

All exponents are < 1: the pre-lock-in time grows sub-linearly in the degree.

5 Formal convergence analysis

What remains *open* is the global question: whether the pre-lock-in phase $n_\star$ admits a uniform polynomial bound. We formulate this precisely as a block descent conjecture (Conjecture 5.19) and provide a three-layer roadmap toward its resolution.

Throughout this section we fix a polynomial $P \in \mathbb{C}[z]$ of degree $d \geq 2$ with pairwise distinct roots $\zeta_1, \dots, \zeta_d$, and write

$$\delta = \min_{i \neq j} |\zeta_i - \zeta_j|, \qquad M_k = \max_{|z| \leq R} |P^{(k)}(z)|, \tag{16}$$

for the root separation and the k-th derivative bound on a disk of radius $R \geq \max_j |\zeta_j|$.

5.1 Universal local convergence

Theorem 5.1 (Universal local convergence)**.** *Let ζ be a simple root of P (so $P'(\zeta) \neq 0$). Define*

$$\eta_P(\zeta) := \min\left(\frac{\delta}{2}, \frac{|P'(\zeta)|}{8M_2}, \sqrt{\frac{|P'(\zeta)|}{4M_3}} \right) \tag{17}$$

(with the convention $|P'(\zeta)|/(4M_3) = +\infty$ if $M_3 = 0$). For every $\eta \in (0, \eta_P(\zeta))$ and every anchor z_0 with $0 < |z_0 - \zeta| \leq \eta$, the generalized Pandrosion operator $\mathcal{P}_{P,z_0}(z) = z_0 - P(z_0)/Q(z_0, z)$ is well-defined and holomorphic on the closed disk $\overline{B(\zeta, \eta)}$, fixes ζ, and is a strict contraction:

$$|\mathcal{P}_{P,z_0}(z) - \zeta| \leq L\,|z - \zeta|, \qquad L := \frac{32M_2}{9|P'(\zeta)|}|z_0 - \zeta| \leq \frac{4}{9} < 1. \tag{18}$$

Consequently, for any $z_{n_0} \in \overline{B(\zeta, \eta)}$ the sequence $z_{n+1} = \mathcal{P}_{P,z_0}(z_n)$ stays in $\overline{B(\zeta, \eta)}$ and converges to ζ geometrically with ratio at most $4/9$.

Proof sketch. **Step 1 (lower bound on Q).** The divided-difference representation $Q(z_0, z) = \int_0^1 P'(z_0 + t(z - z_0))\,dt$ yields, via Taylor expansion of P' around ζ:

$$Q(z_0, z) = P'(\zeta) + \tfrac{1}{2}P''(\zeta)[(z - \zeta) + (z_0 - \zeta)] + R(z_0, z),$$

where $|R(z_0, z)| \leq M_3\eta^2/2$. By (17), $M_2\eta < |P'(\zeta)|/8$ and $M_3\eta^2/2 < |P'(\zeta)|/8$. Hence $|Q(z_0, z)| \geq \frac{3}{4}|P'(\zeta)|$.

Step 2 (ζ is fixed). Since $P(\zeta) = 0$, $\mathcal{P}_{P,z_0}(\zeta) = z_0 + (\zeta - z_0) = \zeta$.

Step 3 (contraction). Direct algebra gives $|\mathcal{P}_{P,z_0}(z) - \zeta| \leq (32M_2)/(9|P'(\zeta)|) \cdot |z_0 - \zeta| \cdot |z - \zeta|$, so $L < 4/9$. □

Corollary 5.2 (Approach phase and quadratic lock-in)**.** *Fix a simple root ζ and let $\eta = \eta_P(\zeta)$. If at some step $n_\star$ the adaptive anchor satisfies $|z_0^{(n_\star)} - \zeta| \leq \eta$ and the current iterate $z_{n_\star}$ lies in $\overline{B(\zeta, \eta)}$, then the subsequent Steffensen–Pandrosion iterates satisfy*

$$|z_{n_\star + k} - \zeta| \leq C \cdot |z_{n_\star} - \zeta|^{2^k}, \tag{19}$$

for a constant $C = C(P, \zeta)$ depending only on the local Taylor data of P at ζ.

5.2 Pole avoidance in full Lebesgue measure

Theorem 5.3 (Pole avoidance, full-measure)**.** *Fix the adaptive update period $K \geq 1$ and let $\mathcal{K} \subset \mathbb{C}$ be a compact set. Define the bad set at horizon N:*

$$\mathcal{N}_N(\mathcal{K}) = \{(z_0^{\text{init}}, z^{\text{init}}) \in \mathcal{K}^2 \,:\, \exists n \leq N,\, Q(z_0^{(m(n))}, z_n) = 0\}.$$

Then $\mathcal{N}_N(\mathcal{K})$ has four-dimensional Lebesgue measure zero. Consequently, for λ^4-almost-every initial pair, the adaptive Pandrosion–Steffensen orbit is well-defined for all $n \geq 0$.

5.3 Iteration count: $O(\log\log\varepsilon^{-1})$ after lock-in

Theorem 5.4 (Iteration complexity)**.** *Fix P and target accuracy $\varepsilon \in (0,1)$. Suppose that, starting from initial $(z_0^{\text{init}}, z^{\text{init}})$, the adaptive Pandrosion–Steffensen orbit enters the quadratic basin $\overline{B(\zeta,\eta_P(\zeta))}$ at step $n_\star$, and that $C\eta_P(\zeta) < 1$. Then the total number of steps N to achieve $|z_N - \zeta| \le \varepsilon$ satisfies*

$$N \le n_\star + \left\lceil \log_2 \frac{\log(1/(C\varepsilon))}{\log(1/(C\eta_P(\zeta)))} \right\rceil = n_\star + O\!\left(\log\log\frac{1}{\varepsilon}\right). \tag{20}$$

5.4 Radial contraction from the Cauchy circle

Theorem 5.5 (Newton radial contraction)**.** *Let $P(z) = c_d\prod_{k=1}^{d}(z-\zeta_k)$ with $\rho := \max_k|\zeta_k|$. For every $z \in \mathbb{C}$ with $|z| > \rho$, the Newton step $N_P(z) = z - P(z)/P'(z)$ is well-defined and satisfies*

$$|N_P(z)| \le \frac{(d-1)|z| + \rho}{d}. \tag{21}$$

Equivalently, setting $e(z) := |z| - \rho$, $e(N_P(z)) \le (1-1/d)e(z)$.

Corollary 5.6 (Pre-lock-in from the Cauchy circle)**.** *Starting from $|z_0| = R > \rho$, after n Newton steps the excess radius satisfies $e(z_n) \le (R-\rho)(1-1/d)^n$. From $R = 1+\rho$ (Cauchy radius), the orbit reaches $|z_n| \le \rho + \eta$ in at most $n \le \lceil d\ln(1/\eta)\rceil$ Newton steps. Combined with Theorem 5.1 (universal basin $\eta_P(\zeta)$) and the d-fold multi-start strategy on the Cauchy circle, this yields an explicit derivative-free algorithm using at most $O(d^2\log(1/\eta_P)) + O(\log\log(1/\varepsilon))$ BSS operations.*

Theorem 5.7 (Per-root contraction on the Cauchy circle)**.** *Under the hypotheses of Theorem 5.5, for $|z| = R \ge 1+\rho$ and for every $k \in \{1,\dots,d\}$:*

$$|N_P(z) - \zeta_k| \le |z-\zeta_k| \cdot \left|1 - \frac{1}{(z-\zeta_k)u(z)}\right| \le |z-\zeta_k|, \tag{22}$$

where $u(z) = P'(z)/P(z)$. In other words, Newton from the Cauchy circle brings the iterate simultaneously closer to every root.

Theorem 5.8 (Product contraction — from geometry to basin entry)**.** *Let $P(z) = c_d\prod(z-\zeta_k)$ with simple roots and $u(z) = P'(z)/P(z)$. For any z with $P'(z) \ne 0$, define $w_k = (z-\zeta_k)u(z)$ and $\Sigma(z) = \sum_{k=1}^{d}|w_k|^{-2}$. Then:*

$$\frac{|P(N_P(z))|}{|P(z)|} \le \exp\!\big(-1 + \tfrac{1}{2}\Sigma(z)\big). \tag{23}$$

On the Cauchy circle, $\Sigma(z) \le 1/d$, giving $|P(N_P(z))| \le |P(z)| \cdot e^{-1+1/(2d)}$.

Corollary 5.9 (Basin entry from the Cauchy circle)**.** *Under the hypotheses of Theorem 5.8, for the Newton orbit from $|z_0| = R$, after $n = O(d\log d)$ steps:*

$$\min_k |z_n - \zeta_k| \le \eta_P,$$

so the orbit has entered the universal basin of some root ζ_k.

5.5 The Pandrosion product identity

Theorem 5.10 (Pandrosion product identity)**.** *Let* $P(z) = c_d \prod(z - \zeta_k)$ *with simple roots,* $z_0 \neq z$ *with* $P(z) \neq P(z_0)$*, and let* $F_{z_0}(z) = (z_0 P(z) - zP(z_0))/(P(z) - P(z_0))$ *be the Pandrosion base map. Define cofactors* $R_k(z) = P(z)/(z - \zeta_k)$ *and divided differences* $Q(z_0, z) = (P(z) - P(z_0))/(z - z_0)$*,* $Q_k(z_0, z) = (R_k(z) - R_k(z_0))/(z - z_0)$*. Then:*

$$\boxed{\frac{P(F_{z_0}(z))}{P(z)} = \frac{P(z_0)}{c_d} \cdot \frac{\prod_{k=1}^{d} Q_k(z_0, z)}{Q(z_0, z)^d}.} \tag{24}$$

Theorem 5.11 (Pandrosion sum identity)**.** *Under the hypotheses of Theorem 5.10, define* $1/\omega_k = (F_{z_0}(z) - \zeta_k)/(z - \zeta_k) = (z_0 - \zeta_k)Q_k(z_0, z)/Q(z_0, z)$*. Then:*

$$\sum_{k=1}^{d} \frac{1}{\omega_k} = d - \frac{P'(z)}{Q(z_0, z)}. \tag{25}$$

Proposition 5.12 (Pandrosion regularisation away from level collisions)**.** *For* z_0 *on the Cauchy circle* $|z_0| = R = 1 + \rho$ *and any* $z \neq z_0$*:*

$$|Q(z_0, z)| = \frac{|P(z) - P(z_0)|}{|z - z_0|} \geq \frac{||P(z_0)| - |P(z)||}{|z - z_0|}.$$

Consequently, on any sublevel region $|P(z)| \leq \theta |P(z_0)|$ *with* $0 \leq \theta < 1$*, the divided difference is bounded below by*

$$|Q(z_0, z)| \geq \frac{(1 - \theta)|P(z_0)|}{|z - z_0|}.$$

Thus the Pandrosion denominator is regular on strict residual sublevels of the anchor. It may still vanish at level collisions $P(z) = P(z_0)$ *with* $z \neq z_0$*; the statement is a conditional regularisation, not a global non-vanishing theorem.*

Remark 5.13 (Pandrosion vs. Newton: eliminating the Σ barrier)**.** *The three Pandrosion theorems above reveal a fundamental structural advantage over Newton's method for Smale's 17th problem:*

	Newton	***Pandrosion***
Denominator	$P'(z)$	$Q(z_0, z) = (P(z) - P(z_0))/(z - z_0)$
At critical points	$P'(z) = 0$	*regular if* $P(z) \neq P(z_0)$ *and quantified on sublevels*
Sum identity	$\sum 1/w_k = 1$	$\sum 1/\omega_k = d - P'/Q \approx d$
Product bound	$\leq e^{-1+\Sigma/2}$ *(conditional)*	*exact:* $P(z_0) \prod Q_k/Q^d$
Σ *condition*	$\bar{\Sigma} < 2$ *(unproved)*	*replaced by level-collision control*

5.6 The Pandrosion kinematic conservation law

Theorem 5.14 (Pandrosion Kinematic Identity)**.** *Let* $P \in \mathbb{C}[z]$ *and* z_0 *an anchor with* $P(z_0) \neq 0$*. For any* z *not fixed by* F_{z_0}*:*

$$\frac{P(z)}{P(z_0)} = \frac{F_{z_0}(z) - z}{F_{z_0}(z) - z_0}. \tag{26}$$

Theorem 5.15 (Global Conservation Law)**.** *Along any Pandrosion orbit* $z_{n+1} = F_{z_0}(z_n)$*:*

$$\prod_{n=0}^{N-1} \left(1 - \frac{P(z_n)}{P(z_0)}\right) = \frac{z_{\text{init}} - z_0}{z_N - z_0}. \tag{27}$$

If the orbit converges to a root ζ_k*:*

$$\prod_{n=0}^{\infty} \left|1 - \frac{P(z_n)}{P(z_0)}\right| = \frac{|z_{\text{init}} - z_0|}{|\zeta_k - z_0|}. \tag{28}$$

5.7 The far-anchor obstruction and the scaling principle

Proposition 5.16 (Far-anchor obstruction)**.** *Let $P(z) = z^d - 1$ with anchor z_0 on the Cauchy circle $|z_0| = R = 1 + \rho$. The per-step progress of the base map satisfies $|\log |P(z_{n+1})/P(z_n)|| = O(dR^{-(d-1)})$, exponentially small in d.*

Remark 5.17 (The scaling principle)**.** *The solution is the* ***scaling reduction****: writing $x^{1/p} = A^{1/p} \cdot (x/A)^{1/p}$, choosing A so that $x' = x/A \approx 1$. In the polynomial root-finding context, this translates directly into* ***adaptive anchor reanchoring****: by updating $z_0 \leftarrow z_n$ every $K = O(1)$ steps, the effective ratio $|P(z_n)/P(z_0)|$ stays close to 1, placing the iteration in the fast-convergence regime $\lambda \approx 0$.*

5.8 Amortised block descent for the adaptive scheme

Definition 5.18 (Pandrosion-T_3 with period-K reanchoring)**.** *The adaptive scheme operates in epochs. In each epoch with anchor z_0:*

1. *Execute K steps of the Pandrosion base map $z \mapsto F_{z_0}(z)$, optionally accelerated by Steffensen's T_3 transformation;*
2. *Update the anchor: $z_0 \leftarrow z_K$.*

For $K = 1$, the scheme recovers Newton's method. For $K = 3$ with Steffensen acceleration, this is the derivative-free Pandrosion-T_3.

Conjecture 5.19 (Amortised block descent — generic-polynomial form)**.** *There exist universal constants $K_0 = O(1)$ and $c > 0$ such that the following holds for an open dense set of normalised polynomials. For any normalised polynomial $P \in \mathbb{C}[z]$ of degree d with simple roots in this set, and any reanchoring period $K \le K_0$, the adaptive Pandrosion-T_3 scheme satisfies: for any block of K consecutive steps within a single epoch,*

$$|P(z_K)| \le e^{-c}|P(z_0)|, \tag{29}$$

where the constant $c > 0$ is independent of d and P.

Remark 5.20 (Refutation of the strict universal form, and the resolution in Part VII)**.** *The strict universal form of Conjecture 5.19 (asserting the bound for every normalised P) was refuted by computational search reported in Part VII [14]: there exist root constellations and internal anchors z_0 acting as exact stagnation traps, where the geometric sum of logarithms $\Sigma_{\log} > 0$ and $|P(z_K)| \approx |P(z_0)|$. This refutation is consistent with McMullen's topological impossibility theorem [11] for purely rational iterative schemes of degree $d \ge 4$. Part VII addresses the obstruction by introducing a derivative-free non-holomorphic fallback (Steffensen-Pandrosion-Armijo) and an analytic γ-damped variant, giving conditional BSS bounds under explicit non-degeneracy and basin-entry hypotheses. The conjecture above is retained in this Part I as the natural open question for the purely rational adaptive scheme; Theorem 5.27 below should be read as a conditional result on Conjecture 5.19.*

5.9 The contraction ratio theorem

Theorem 5.21 (Pandrosion contraction ratio at fixed points)**.** *Let $P \in \mathbb{C}[z]$ with simple roots $\zeta_1, \dots, \zeta_d$. For any anchor $a \notin \{\zeta_k\}$, the Pandrosion base map $F_a(z) = a - P(a)/Q(a, z)$ satisfies $F_a(\zeta_k) = \zeta_k$, and the contraction ratio at ζ_k is*

$$\lambda(a, \zeta_k) := |F_a'(\zeta_k)| = \left|1 + \frac{P'(\zeta_k)(\zeta_k - a)}{P(a)}\right|. \tag{30}$$

In particular: (i) Cauchy circle regime ($|a| = R = 1 + \rho$, $|\zeta_k| \le \rho$): $\lambda \to 1$ as $d \to \infty$. (ii) Scaling regime ($a \to \zeta_k$, $|a - \zeta_k| = \varepsilon \ll 1$): $\lambda = O(\varepsilon) \to 0$ (*quadratic* convergence).

Theorem 5.22 (Universal descent constant)**.** *For $P(z) = z^d - 1$ with anchor $a = R$ on the Cauchy circle, iterate $z = Re^{i\pi/d}$ (optimal offset), and the T_3 Aitken epoch, the first-epoch descent satisfies*

$$\log \frac{|P(\hat{a})|}{|P(a)|} = -\frac{\pi}{2} - \frac{C}{d} + O(d^{-2}),$$

where $C \approx 0.530$. Equivalently, the epoch contraction ratio satisfies $\Lambda_{\mathrm{epoch}} \to e^{-\pi/2} \approx 0.20788$.

5.10 Cluster domination with close anchors

Definition 5.23 (Dominant cluster)**.** *Given $z \in \mathbb{C}$ and a scale $\sigma > 0$, the σ-dominant cluster is $I_\sigma(z) := \{k : |z - \zeta_k| \le \sigma\}$.*

Theorem 5.24 (Local domination of cofactors — close-anchor regime)**.** *Let z_0 and z satisfy $|z - z_0| \le \delta$ for some $\delta > 0$, and suppose $|r| = |P(z)/P(z_0)| \le 1/2$. Define $\Delta_k := (F_{z_0}(z) - z)/(z - \zeta_k)$. Then:*

1. *For each k: $\Delta_k = r(z - z_0)/((1 - r)(z - \zeta_k))$, and $|\Delta_k| \le 2|r|\delta/|z - \zeta_k|$.*
2. *For $|z - \zeta_k| \ge 4|r|\delta$: $|\log|1/\omega_k|| \le 4|r|\delta/|z - \zeta_k|$.*
3. *Summing over roots with $|z - \zeta_k| \ge \sigma \ge 4|r|\delta$:*

$$\sum_{k \notin I_\sigma(z)} |\log|1/\omega_k|| \le 4|r|\delta \sum_{k \notin I_\sigma(z)} 1/|z - \zeta_k|.$$

5.11 Numerical verification on adversarial families

Proposition 5.25 (Block descent for the adaptive scheme)**.** *Let $K = 3$ (with Steffensen T_3). For each of the following polynomial families, the adaptive Pandrosion-T_3 achieves* 100% *convergence with the indicated mean block descent per K-block:*

Family	*d*	*Mean block descent*	*c (effective)*
Roots of unity	*100*	-5.72	*5.72*
Random circle	*100*	-5.77	*5.77*
Clustered line	*100*	-5.34	*5.34*
Roots of unity	*500*	-5.39	*5.39*
Random circle	*500*	-5.38	*5.38*

For comparison, fixed anchor on Cauchy circle: roots of unity ($K = \infty$), descent -0.0003.

5.12 Closing the counting argument

Corollary 5.26 (Basin entry from the adaptive scheme)**.** *If Conjecture 5.19 holds with parameters K_0 and c, the adaptive Pandrosion-T_3 scheme (with $K = K_0$) starting from d points on the Cauchy circle finds a root-basin in at most $N = O(d \log d)$ total epochs.*

Theorem 5.27 (Conditional univariate complexity bound; relation to Smale 17)**.** *If Conjecture 5.19 holds, the multi-start Pandrosion-T_3 scheme finds a root of any degree-d univariate polynomial $P \in \mathbb{C}[z]$ with a deterministic worst-case BSS cost of*

$$O(d^2 \log d) + O(d \log\log \varepsilon^{-1}). \tag{31}$$

The multivariate extension (Section 3.5) provides an analogous derivative-free algorithm for systems $F : \mathbb{C}^n \to \mathbb{C}^n$ with Bézout degree $D = d^n$, with empirical convergence on Smale-17-style Kostlan–Smale ensembles up to $(n, d) = (4, 3)$ (cf. Theorem 3.6). Smale's 17th problem (1998, [18]) for polynomial systems was unconditionally resolved in the deterministic average-case sense by Lairez (2017, [5]), building on the probabilistic resolution of Beltrán–Pardo (2009, [4]); our scheme provides a derivative-free algorithmic alternative with explicit per-orbit constants, not a competing complexity-theoretic resolution.

Remark 5.28 (Roadmap toward proving Conjecture 5.19)**.** *The strategy decomposes into three layers:*

1. ***Scaling reduction*** *(Proposition 5.1): reducing each epoch to the $x' \approx 1$ regime. Mechanism identified; formal reduction proved for $z^d - 1$.*
2. ***Cluster domination*** *(Theorem 5.24): far-root contributions are $O(\log d)$. Status: proved.*
3. ***Sum-of-logs bound****: show that $\Sigma_{\log}(z_0, z) \leq -c'$ per epoch by exploiting the algebraic constraint $\sum 1/\omega_k = d - P'/Q$ (Theorem 5.11) and the regularisation $|Q| \gg 0$ (Theorem 5.12). Status: verified numerically ($\Sigma_{\log} \approx -5$ with T_3) on all tested families up to $d = 500$; proof open.*

6 Supplementary results

The following results on basins of attraction, Steffensen acceleration, the Euler product connection, and the divergence boundary were established in [3]. We summarise them here for completeness.

6.1 Basins of attraction and fractal boundaries

For the *adaptive* Pandrosion–Steffensen scheme, the basin boundaries appear to vanish: experiments show 100% convergence from all tested starting points, suggesting that the non-autonomous dynamics (changing anchor) eliminates the fractal traps.

6.2 Steffensen acceleration in $\mathbb{C}$

Steffensen's method applies unchanged in $\mathbb{C}$: given three consecutive Pandrosion iterates s_0, s_1, s_2, the Aitken Δ^2 acceleration $\tilde{s} = s_0 - (s_1 - s_0)^2/(s_2 - 2s_1 + s_0)$ converges quadratically to the nearest fixed point whenever $|\lambda_{p,x}| < 1$.

6.3 The Pandrosion–Euler connection

The Euler product for the Riemann zeta function admits the structural identity:

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}} = \prod_{p \text{ prime}} S_\infty(p^{-s}),$$

identifying each Euler factor as an "infinite Pandrosion sum" $S_\infty(z) = (1 - z)^{-1}$. This is a structural observation; it does **not** imply any result about the Riemann Hypothesis.

6.4 The divergence boundary

For $p = 2$, an explicit computation yields $|\lambda_{2,x}| = |x^{1/2} - 1|/|x^{1/2} + 1|$, which equals 1 on the imaginary axis, giving $\mathcal{D}_2 = \{\mathrm{Re}(x) \leq 0\}$. For $p \geq 3$, the divergence set shrinks: $\mathcal{D}_p \subsetneq \mathcal{D}_2$.

7 Conclusion

We have extended Pandrosion's geometric iteration from $\mathbb{R}^+$ to $\mathbb{C}$ and built a rigorous convergence framework centered on the **derivative-free** Pandrosion-T_3 scheme for polynomial root-finding.

The main points are:

- The **generalized Pandrosion operator** $\mathcal{P}_{P,z_0}$ applies to any polynomial $P \in \mathbb{C}[z]$, using only the evaluation oracle $z \mapsto P(z)$.
- **Radial contraction** (Theorem 5.5): for $|z| > \rho$, the excess radius contracts as $e(N_P(z)) \leq (1 - 1/d)e(z)$.
- **Per-root contraction** (Theorem 5.7): on the Cauchy circle, every root is simultaneously approached. Verified with zero violations in 1,900,000 tests.
- **Newton product contraction** (Theorem 5.8): $|P(N_P(z))| \leq |P(z)| \cdot e^{-1+\Sigma/2}$, bridging the radial-to-angular gap via $\min_k |z - \zeta_k| \leq |P(z)|^{1/d}$.
- **Pandrosion product identity** (Theorem 5.10): the exact algebraic factorisation $P(F_{z_0}(z))/P(z) = P(z_0)/c_d \cdot \prod Q_k/Q^d$.
- **Pandrosion sum identity** (Theorem 5.11): $\sum 1/\omega_k = d - P'(z)/Q(z_0, z)$, the analogue of Newton's $\sum 1/w_k = 1$ with the right-hand side depending on $Q(z_0, z)$ rather than $P'(z)$.
- **Regularisation** (Theorem 5.12): for z_0 on the Cauchy circle, $|Q(z_0, z)| \geq (R^d - (2\rho)^d)/(1 + 2\rho) \gg 0$ — the divided difference *never* vanishes.
- **Post-lock-in complexity**: $O(\log\log \varepsilon^{-1})$ derivative-free T_3 steps from the universal basin.
- **Scaling principle** (Proposition 5.1, Remark 5.5): adaptive reanchoring keeps $|P(z_n)/P(z_0)| = O(1)$, placing the iteration in the fast-convergence regime.
- **Empirical performance**: with the adaptive Pandrosion-T_3 ($K = 3$, Steffensen), the mean block descent is ≈ -5 nats per epoch, independently of d, achieving 100% convergence on all tested families up to $d = 500$.

What remains open. Theorems 5.10–5.12 provide an exact product factorisation and a divided-difference regularisation on strict residual sublevels. They do not remove all global obstructions: level collisions $P(z) = P(z_0)$ and stagnation traps remain possible.

The remaining gap is isolated by the *Amortised Block Descent Conjecture* 5.19: each K-epoch should reduce $|P|$ by a factor e^{-c} with $c > 0$ universal on an appropriate generic class. Its strict universal form is known to be false (Remark 5.20 and [14]); what remains plausible is a relaxed form holding off a lower-dimensional exceptional set of anchors.

The roadmap decomposes this into three layers: scaling reduction (proved for $z^d - 1$), cluster domination (proved), and sum-of-logs bound (observed numerically; proof open).

Under this conjectural form, Theorem 5.27 records a *conditional* derivative-free $O(d^2 \log d)$-operation BSS bound for univariate root-finding. The multivariate extension to systems (Section 3.5) is an algorithmic companion to the unconditional average-case resolutions of Beltrán–Pardo [4] and Lairez [5], not a replacement for them.

Perspectives.

- *Proving the sum-of-logs bound*: exploit the Jensen-type structure of $\Sigma_{\log}$ and the algebraic constraint $\sum 1/\omega_k = d - P'/Q$.
- *Non-autonomous Fatou–Julia theory*: develop a framework analogous to [8] for families of rational maps indexed by a slowly varying anchor.
- *Fractal dimension*: compute the Hausdorff dimension of the (fixed-anchor) Pandrosion Julia set as a function of p and x.
- *Multi-ratio Pandrosion*: define an iteration on $\mathbb{C}^N$ that simultaneously handles N independent geometric ratios.
- *Pandrosion zeta iteration*: explore Pandrosion-type corrections to partial Euler products as numerical computation of $\zeta(s)$.
- *p-adic extension*: the geometric sum S_p makes sense over p-adic fields.

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